

Technical Appendix to “Credit and Banking in a DSGE Model of the Euro Area”

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In this Appendix we include additional material concerning both the model and the estimation. First, we briefly present the model and report the full set of equations. Second we discuss the data and provide the details on their construction. Third we describe the estimation methodology. Finally we study the convergence of the Metropolis algorithm and check the robustness of the results along some important dimensions.

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1 The model

In this Section we summarize the equations describing the equilibrium of the model.

1.1 Patient households

The representative patient household i maximizes the expected utility

$$E_0 \sum_{t=0}^{\infty} \beta_P^t \left[(1 - a^P) \varepsilon_t^z \log(c_t^P(i) - a^P c_{t-1}^P) + \varepsilon_t^h \log h_t^P(i) - \frac{l_t^P(i)^{1+\phi}}{1+\phi} \right]$$

subject to the budget constraint

$$c_t^P(i) + q_t^h \Delta h_t^P(i) + d_t^P(i) \leq w_t^P l_t^P(i) + (1 + r_{t-1}^d) d_{t-1}^P(i) / \pi_t + t_t^P(i)$$

The choice variables for the patient household are c_t^P , h_t^P and d_t^P . The first order conditions are (omitting i 's):

$$\begin{aligned} \lambda_t^P &= \varepsilon_t^z (1 - a^P) / (c_t^P - a^P c_{t-1}^P) \\ \lambda_t^P q_t^h &= \frac{\varepsilon_t^h}{h_t^P} + \beta^P E_t [\lambda_{t+1}^P q_{t+1}^h] \\ \lambda_t^P &= \beta^P E_t [\lambda_{t+1}^P (1 + r_t^d) / \pi_{t+1}] \end{aligned}$$

where λ_t^P is the multiplier on the budget constraint expressed in real terms.

1.2 Impatient households

The representative impatient household i maximizes the expected utility

$$E_0 \sum_{t=0}^{\infty} \beta_I^t \left[(1 - a^I) \varepsilon_t^z \log(c_t^I(i) - a^I c_{t-1}^I) + \varepsilon_t^h \log h_t^I(i) - \frac{l_t^I(i)^{1+\phi}}{1+\phi} \right]$$

subject to the budget constraint (associated to a Lagrange multiplier λ_t^I)

$$c_t^I(i) + q_t^h \Delta h_t^I(i) + (1 + r_{t-1}^{bH}) b_{t-1}^I(i) / \pi_t \leq w_t^I l_t^I(i) + b_t^I(i) + t_t^I(i)$$

and to a borrowing constraint (with multiplier s_t^I)

$$(1 + r_t^{bH}) b_t^I(i) \leq m_t^I E_t [q_{t+1}^h h_{t+1}^I(i) \pi_{t+1}]$$

The choice variables for the impatient household are c_t^I , h_t^I and b_t^I . The first-order conditions of the patient household's maximization problem are

$$\begin{aligned} \lambda_t^I &= \varepsilon_t^z (1 - a^I) / (c_t^I - a^I c_{t-1}^I) \\ \lambda_t^I q_t^h &= \frac{\varepsilon_t^h}{h_t^I} + \beta^I E_t [\lambda_{t+1}^I q_{t+1}^h + s_t^I m_t^I q_{t+1}^h \pi_{t+1}] \\ \lambda_t^I &= \beta^I E_t [\lambda_{t+1}^I (1 + r_t^d) / \pi_{t+1}] \end{aligned}$$

plus the budget constraint and the borrowing constraint (solved with equality). λ_t^I is the multiplier of the budget constraint expressed in real terms.

1.3 Entrepreneurs

Each entrepreneur i maximizes the utility function

$$E_0 \sum_{t=0}^{\infty} \beta_E^t \log(c_t^E(i) - a^E c_{t-1}^E)$$

subject to the budget constraint (with multiplier λ_t^E)

$$c_t^E(i) + w_t^P l_t^{E,P}(i) + w_t^I l_t^{E,I}(i) + \frac{1+r_t^{bE}}{\pi_t} b_{t-1}^E(i) + q_t^k k_t^E(i) + \psi(u_t(i)) k_{t-1}^E(i) \leq \frac{y_t^E(i)}{x_t} + b_t^E(i) + q_t^k (1-\delta) k_{t-1}^E(i)$$

with $\psi(u_t) = \xi_1(u_t - 1) + \frac{\xi_2}{2}(u_t - 1)^2$ (following Schmitt-Grohé and Uribe, 2006). His available technology has

$$y_t^E(i) = a_t^E [k_{t-1}^E(i) u_t(i)]^\alpha l_t^E(i)^{1-\alpha}$$

where aggregate labor is a combination of inputs from patient and impatient households according to $l_t^E = (l_t^{E,P})^\mu (l_t^{E,I})^{1-\mu}$, with parameter μ calibrated to measure the labor income share of unconstrained households as in Iacoviello and Neri (2009). The entrepreneur faces a borrowing constraint of the kind

$$(1 + r_t^{bE}) b_t^E(i) \leq m_t^E E_t(q_{t+1}^k \pi_{t+1} (1 - \delta) k_t^E(i))$$

Entrepreneur's choice variables are c_t^E , k_t^E , u_t , $l_t^{E,P}$, $l_t^{E,I}$ and b_t^E . Optimality conditions include the budget constraint, an always binding borrowing constraint and FOCs:

$$\begin{aligned} \lambda_t^E &= (1 - a^E)/(c_t^E - a^E c_{t-1}^E) \\ \lambda_t^E q_t^k &= E_t \{ s_t^E m_t^E q_{t+1}^k \pi_{t+1} (1 - \delta) + \beta_E \lambda_{t+1}^E [r_{t+1}^k u_{t+1} + q_{t+1}^k (1 - \delta) - \psi(u_{t+1})] \} \\ r_t^k &= \xi_1 + \xi_2 (u_t - 1) \\ w_t^P &= (1 - \alpha) \frac{y_t^E}{x_t} \frac{\mu}{l_t^{E,P}} \\ w_t^I &= (1 - \alpha) \frac{y_t^E}{x_t} \frac{1 - \mu}{l_t^{E,I}} \\ \lambda_t^E &= s_t^E (1 + r_t^{bE}) + \beta_E E_t [\lambda_{t+1}^E (1 + r_t^{bE}) / \pi_{t+1}] \end{aligned}$$

where $r_t^k \equiv \alpha a_t^E [k_{t-1}^E u_t]^\alpha l_t^E(i)^{1-\alpha} / x_t$.

1.4 Capital goods producers

CGP firms solve the following problem

$$\max_{\{\bar{x}_t, i_t\}} E_0 \sum_{t=0}^{\infty} \Lambda_{0,t}^E (q_t^k \Delta \bar{x}_t - i_t)$$

subject to

$$\bar{x}_t = \bar{x}_{t-1} + \left[1 - \frac{\kappa_i}{2} \left(\frac{i_t \varepsilon_t^{qk}}{i_{t-1}} - 1 \right)^2 \right] i_t$$

where $\Delta \bar{x}_t = k_t - (1 - \delta)k_{t-1}$ is the flow output.

From FOCs, the amount of new capital that CGP firms can produce is given by¹

$$k_t = (1 - \delta)k_{t-1} + \left[1 - \frac{\kappa_i}{2} \left(\frac{i_t \varepsilon_t^{qk}}{i_{t-1}} - 1 \right)^2 \right] i_t$$

and the real price of capital q_t^k is determined by

$$1 = q_t^k \left[1 - \frac{\kappa_i}{2} \left(\frac{i_t \varepsilon_t^{qk}}{i_{t-1}} - 1 \right)^2 - \kappa_i \left(\frac{i_t \varepsilon_t^{qk}}{i_{t-1}} - 1 \right) \frac{i_t \varepsilon_t^{qk}}{i_{t-1}} \right] + \beta_E E_t \left[\frac{\lambda_{t+1}^E}{\lambda_t^E} q_{t+1}^k \varepsilon_{t+1}^{qk} \kappa_i \left(\frac{i_{t+1} \varepsilon_{t+1}^{qk}}{i_t} - 1 \right) \left(\frac{i_{t+1}}{i_t} \right)^2 \right]$$

1.5 Labor market

1.5.1 Labor packers

A perfectly competitive labor packer $s \in \{P, I\}$ demands differentiated labor services from unions (s, m) ($m \in [0, 1]$) in order to assemble them into a CES composite labor input to be supplied to entrepreneurs. He solves

$$\begin{aligned} \max_{l_t^s(m)} l_t^s &= \left[\int_0^1 l_t^s(m)^{\frac{\varepsilon^l - 1}{\varepsilon^l}} dm \right]^{\frac{\varepsilon^l}{\varepsilon^l - 1}} \\ \text{s.t. } &\int_0^1 W_t^s(m) l_t^s(m) dm \leq \bar{E}_t \end{aligned}$$

for a given level of the overall wage bill $\bar{E}_t$. The solution gives the demand for each kind of differentiated labor service $l_t^s(m)$:

$$l_t^s(m) = \left(\frac{W_t^s(m)}{W_t^s} \right)^{-\varepsilon^l} l_t^s$$

¹ As pointed out by BGG (1999), a totally equivalent expression for the price of capital could be obtained by internalizing the capital formation problem within the entrepreneurs' problem.

where

$$W_t^s = \left[\int_0^1 W_t^s(m)^{1-\varepsilon^l} dm \right]^{\frac{1}{1-\varepsilon^l}}$$

1.5.2 Unions

Workers (i) sell their slightly differentiated labor types through unions, which for each labor type m exist in two kinds: one for patient and one for impatient households (indexed by $s \in \{P, I\}$). Each union (s, m) sets nominal wages $\{W_t^s(m)\}_{t=0}^\infty$ by maximizing

$$E_0 \sum_{t=0}^\infty \beta_s^t \left\{ U_{c_t^s(i,m)} \left[\frac{W_t^s(m)}{P_t} l_t^s(i, m) - \frac{\kappa_w}{2} \left(\frac{W_t^s(m)}{W_{t-1}^s(m)} - \pi_{t-1}^{\iota_w} \pi^{1-\iota_w} \right)^2 \frac{W_t^s}{P_t} \right] - \frac{l_t^s(i, m)^{1+\phi}}{1+\phi} \right\}$$

subject to demand

$$l_t^s(i, m) = l_t^s(m) = \left(\frac{W_t^s(m)}{W_t^s} \right)^{-\varepsilon^l} l_t^s$$

In a symmetric equilibrium, the labor choice for a household of type s will be given by an ensuing (non-linear) wage-Phillips curve

$$\kappa_w (\pi_t^{w^s} - \pi_{t-1}^{\iota_w} \pi^{1-\iota_w}) \pi_t^{w^s} = \beta_s E_t \left[\frac{\lambda_{t+1}^s}{\lambda_t^s} \kappa_w (\pi_{t+1}^{w^s} - \pi_t^{\iota_w} \pi^{1-\iota_w}) \frac{\pi_{t+1}^{w^s 2}}{\pi_{t+1}} \right] + (1 - \varepsilon^l) l_t^s + \frac{\varepsilon^l l_t^{s1+\phi}}{w_t^s \lambda_t^s}$$

where ω_t^s is the real wage and nominal type s wage inflation is equal to

$$\pi_t^{w^s} = \frac{w_t^s}{w_{t-1}^s} \pi_t$$

1.6 Loans and deposits demand

We assume that units of deposits and of loan contracts bought by households and entrepreneurs are a composite CES basket of slightly differentiated products that each bank j supplies.

Household i seeking an amount of real loans equal to $\bar{b}_t^I(i)$ minimizes the due total repayment

$$\min_{\{b_t^I(i,j)\}} \int_0^1 r_t^{bH}(j) b_t^I(i, j) dj$$

subject to

$$\left[\int_0^1 b_t^I(i, j)^{\frac{\varepsilon_t^{bH}-1}{\varepsilon_t^{bH}}} dj \right]^{\frac{\varepsilon_t^{bH}}{\varepsilon_t^{bH}-1}} \geq \bar{b}_t^I(i) .$$

with $\varepsilon_t^{bH} > 1$ for all t . Combining FOCs across all impatient households, aggregate impatient households' demand for loans at bank j is obtained as

$$b_t^I(j) = \left(\frac{r_t^{bH}(j)}{r_t^{bH}} \right)^{-\varepsilon_t^{bH}} b_t^I \quad (1)$$

where b_t^I is aggregate demand for household loans and the interest rates index for loans to households is $r_t^{bH} = [\int_0^1 r_t^{bH}(j)^{1-\varepsilon_t^{bH}} dj]^{\frac{1}{1-\varepsilon_t^{bH}}}$.

Demand for entrepreneurs' loans is obtained analogously and is equal to

$$b_t^E(j) = \left(\frac{r_t^{bE}(j)}{r_t^{bE}} \right)^{-\varepsilon_t^{bE}} b_t^E \quad (2)$$

Demand for deposits at bank j of impatient household i , seeking an overall amount of real savings $\bar{d}_t^P(i)$, is obtained by maximizing the revenue of total savings

$$\max_{\{d_t(i,j)\}} \int_0^1 r_t^d(j) d_t^P(i, j) dj$$

subject to the aggregation technology (with $\varepsilon_t^d < -1$ for all t)

$$\left[\int_0^1 d_t^P(i, j)^{\frac{\varepsilon_t^d - 1}{\varepsilon_t^d}} dj \right]^{\frac{\varepsilon_t^d}{\varepsilon_t^d - 1}} \leq \bar{d}_t^P(i) .$$

Combining FOCs, aggregate households' demand for deposits at bank j , $d_t^P(j)$, is given by

$$d_t^P(j) = \left(\frac{r_t^d(j)}{r_t^d} \right)^{-\varepsilon_t^d} d_t \quad (3)$$

where d_t is aggregate deposits demand and $r_t^d = [\int_0^1 r_t^d(j)^{1-\varepsilon_t^d} dj]^{\frac{1}{1-\varepsilon_t^d}}$ is the deposit rate index.

1.7 Banks

1.7.1 Wholesale branch

The problem for the wholesale bank is

$$\begin{aligned} \max_{\{B_t, D_t\}} E_0 \sum_{t=0}^{\infty} \Lambda_{0,t}^P & \left[(1 + R_t^b) B_t - B_{t+1} \pi_{t+1} + D_{t+1} \pi_{t+1} - (1 + R_t^d) D_t + (K_{t+1}^b \pi_{t+1} - K_t^b) - \frac{\kappa_{Kb}}{2} \left(\frac{K_t^b}{B_t} - \nu^b \right)^2 K_t^b \right] \\ \text{s.t.} \quad & B_t = D_t + K_t^b \end{aligned}$$

The FOCs deliver

$$R_t^b = R_t^d - \kappa_{Kb} \left(\frac{K_t^b}{B_t} - \nu^b \right) \left(\frac{K_t^b}{B_t} \right)^2 \quad (4)$$

No arbitrage with a lending facility at the central bank ensures $R_t^d \equiv r_t$ and therefore (4) becomes

$$R_t^b = r_t - \kappa_{Kb} \left(\frac{K_t^b}{B_t} - \nu^b \right) \left(\frac{K_t^b}{B_t} \right)^2$$

1.7.2 Retail banking

Retail banks are monopolistic competitors both on the loans and on the deposits markets.

Loan branch: The retail loan branch of bank $j \in [0, 1]$ solves

$$\begin{aligned} \max_{\{r_t^{bH}(j), r_t^{bE}(j)\}} E_0 \sum_{t=0}^{\infty} \Lambda_{0,t}^P & \left[r_t^{bH}(j) b_t^I(j) + r_t^{bE}(j) b_t^E(j) - R_t^b B_t(j) \right. \\ & \left. - \frac{\kappa_{bH}}{2} \left(\frac{r_t^{bH}(j)}{r_{t-1}^{bH}(j)} - 1 \right)^2 r_t^{bH} b_t^I - \frac{\kappa_{bE}}{2} \left(\frac{r_t^{bE}(j)}{r_{t-1}^{bE}(j)} - 1 \right)^2 r_t^{bE} b_t^E \right] \end{aligned}$$

subject to demand schedules (1)-(2) and with $B_t(j) = b_t(j) = b_t^I(j) + b_t^E(j)$. For rates to households and firms (indexed by s), the first order conditions yield, after imposing a symmetric equilibrium,

$$1 - \varepsilon_t^{bs} + \varepsilon_t^{bs} \frac{R_t^b}{r_t^{bs}} - \kappa_{bs} \left(\frac{r_t^{bs}}{r_{t-1}^{bs}} - 1 \right) \frac{r_t^{bs}}{r_{t-1}^{bs}} + \beta_P E_t \left\{ \frac{\lambda_{t+1}^P}{\lambda_t^P} \kappa_{bs} \left(\frac{r_{t+1}^{bs}}{r_t^{bs}} - 1 \right) \left(\frac{r_{t+1}^{bs}}{r_t^{bs}} \right)^2 \frac{b_{t+1}^s}{b_t^s} \right\} = 0$$

Deposit branch: The retail deposit branch of bank j solves

$$\max_{\{r_t^d(j)\}} E_0 \sum_{t=0}^{\infty} \Lambda_{0,t}^P \left[r_t D_t(j) - r_t^d(j) d_t^P(j) - \frac{\kappa_d}{2} \left(\frac{r_t^d(j)}{r_{t-1}^d(j)} - 1 \right)^2 r_t^d d_t \right]$$

subject to deposits demand (3) and with $D_t(j) = d_t^P(j)$. After imposing a symmetric equilibrium, the first-order condition for optimal deposit interest rate setting is

$$-1 + \varepsilon_t^d - \varepsilon_t^d \frac{r_t}{r_t^d} - \kappa_d \left(\frac{r_t^d}{r_{t-1}^d} - 1 \right) \frac{r_t^d}{r_{t-1}^d} + \beta_P E_t \left\{ \frac{\lambda_{t+1}^P}{\lambda_t^P} \kappa_d \left(\frac{r_{t+1}^d}{r_t^d} - 1 \right) \left(\frac{r_{t+1}^d}{r_t^d} \right)^2 \frac{d_{t+1}}{d_t} \right\} = 0 .$$

Overall bank profits are the sum of net earnings (intermediation margins minus other costs) from the wholesale unit and the retail branches. After deleting intra-group transactions, their real terms expression reads

$$j_t^b = r_t^{bH} b_t^H + r_t^{bE} b_t^E - r_t^d d_t - \frac{\kappa_{Kb}}{2} \left(\frac{K_t^b}{B_t} - \nu^b \right)^2 K_t^b - Adj_t^B$$

where Adj_t^B indicates adjustment costs for changing interest rates on loans and deposits.

Bank capital is accumulated out of retained earnings according to (in real terms):

$$K_t^b = (1 - \delta^b) \frac{K_{t-1}^b}{\pi_t} + \frac{j_{t-1}^b}{\pi_t}$$

1.8 Goods retailers

The retail goods market is assumed to be monopolistically competitive. Retailers solve the following problem

$$\max_{\{P_t(j)\}} E_0 \sum_{t=0}^{\infty} \Lambda_{0,t}^P \left[P_t(j) y_t(j) - P_t^W y_t(j) - \frac{\kappa_p}{2} \left(\frac{P_t(j)}{P_{t-1}(j)} - \pi_{t-1}^{\iota_p} \pi^{1-\iota_p} \right)^2 P_t y_t \right]$$

subject to a demand coming from consumers maximization of a consumption aggregator

$$y_t(j) = \left(\frac{P_t(j)}{P_t} \right)^{-\varepsilon_t^y} y_t$$

In a symmetric equilibrium, FOCs give rise to a non-linear Phillips curve

$$1 - \varepsilon_t^y + \frac{\varepsilon_t^y}{x_t} - \kappa_p (\pi_t - \pi_{t-1}^{\iota_p} \pi^{1-\iota_p}) \pi_t + \beta_P E_t \left[\frac{\lambda_{t+1}^P}{\lambda_t^P} \kappa_p (\pi_{t+1} - \pi_t^{\iota_p} \pi^{1-\iota_p}) \pi_{t+1} \frac{y_{t+1}}{y_t} \right] = 0$$

Aggregate retailers' profits are instead given by

$$j_t^R = y_t \left[1 - \frac{1}{x_t} - \frac{\kappa_p}{2} (\pi_t - \pi_{t-1}^{\iota_p} \pi^{1-\iota_p})^2 \right]$$

1.9 Monetary policy

The central bank sets the policy rate r_t according to a Taylor rule of the type

$$(1 + r_t) = (1 + r)^{(1-\phi_R)} (1 + r_{t-1})^{\phi_R} \left(\frac{\pi_t}{\pi} \right)^{\phi_\pi(1-\phi_R)} \left(\frac{y_t}{y_{t-1}} \right)^{\phi_y(1-\phi_R)} \varepsilon_t^r$$

1.10 Aggregation and market clearing

In the final good market, the equilibrium condition is given by the following resource constraint

$$y_t = c_t + q_t^k [k_t - (1 - \delta) k_{t-1}] + k_{t-1} \psi(u_t) + \delta^b \frac{K_{t-1}^b}{\pi_t} + Adj_t$$

where $c_t = c_t^P + c_t^I + c_t^E$ and $k_t = \gamma^E k_t^E(i)$ (γ^s , $s \in P, I, E$, is the measure of each subset of agents). Equilibrium in the housing market is given by

$$\bar{h} = \gamma^P h_t^P(i) + \gamma^I h_t^I(i)$$

2 Data and sources

Real consumption: Consumption of households and non-profit institutions serving households (NPISH), constant prices, seasonally adjusted, not working day adjusted, euro area 15 (Eurostat).

Real investment: Gross fixed capital formation, constant prices, seasonally adjusted, not working day adjusted, euro area 15 (Eurostat).

Real house prices: Nominal residential property prices deflated with the harmonized index of consumer prices (ECB and Eurostat).

Wages: Hourly labor cost index - wages and salaries, whole economy excluding agriculture, fishing and government sectors, seasonally and working day adjusted (Eurostat).

Inflation: HICP overall index, quarterly changes, seasonally adjusted, not working day adjusted, euro area 15 (ECB).

Nominal interest rate (policy): Eonia rate (ECB).

Interest rate on loans to households: Annualized agreed rate (AAR) on loans for house purchases, total maturity, new business coverage (ECB).

Interest rate on loans to firms: AAR on loans other than bank overdrafts to non-financial corporations with maturity of over one year, new business coverage (ECB).

Interest rate on deposits: Weighted average (with weights proportional to outstanding amounts) of AARs on overnight deposits (total maturity), on deposits with agreed maturity of up to two years, and on deposits redeemable at notice of up to three months, households and non-profit institutions serving households, new business coverage (ECB).

Loans to households: Outstanding amounts of loans to households for house purchasing, total maturity, neither seasonally nor working day adjusted (ECB).

Loans to firms: Outstanding amounts of loans to non-financial corporations, total maturity, neither seasonally nor working day adjusted (ECB).

Deposits: Overnight, with agreed maturity up to 2 years, redeemable at notice up to 3 months; outstanding amounts; households and NPISH (ECB).

For bank retail interest rates, we merged two ECB data sets. From 2003M1, we use harmonized monthly data from the MFI Interest Rate (MIR) statistics in new business coverage as it reflects better than the outstanding amounts the extent to which changes in the policy rate are transmitted to bank rates. Data from MIR are extended back to 1998M1 using the euro area Retail Interest Rate (RIR) data set, compiled by the ECB until 2003M9. Since original national data in RIR are neither harmonized in coverage nor in nature (nominal or effective), we check the stability of the relation between comparable

MIR-RIR rates series over the nine-month overlapping period of 2003 before using monthly variations in comparable series from RIR to backcast MIR rates.

Volumes of loans and deposits refer to outstanding amounts: if data on new businesses were used (available from 2003Q1 in MIR) the high volatility of these series would, first, not allow a safe backcasting of volumes data and, second, induce undue instability when used as weights for aggregating the interest rate series.

Figure A1 reports the raw data while figure A2 the transformed time series used in the estimation of the model. Figure A3 reports monthly observations on retail bank rates and the overnight rate (EONIA).

Figure A1. Raw data

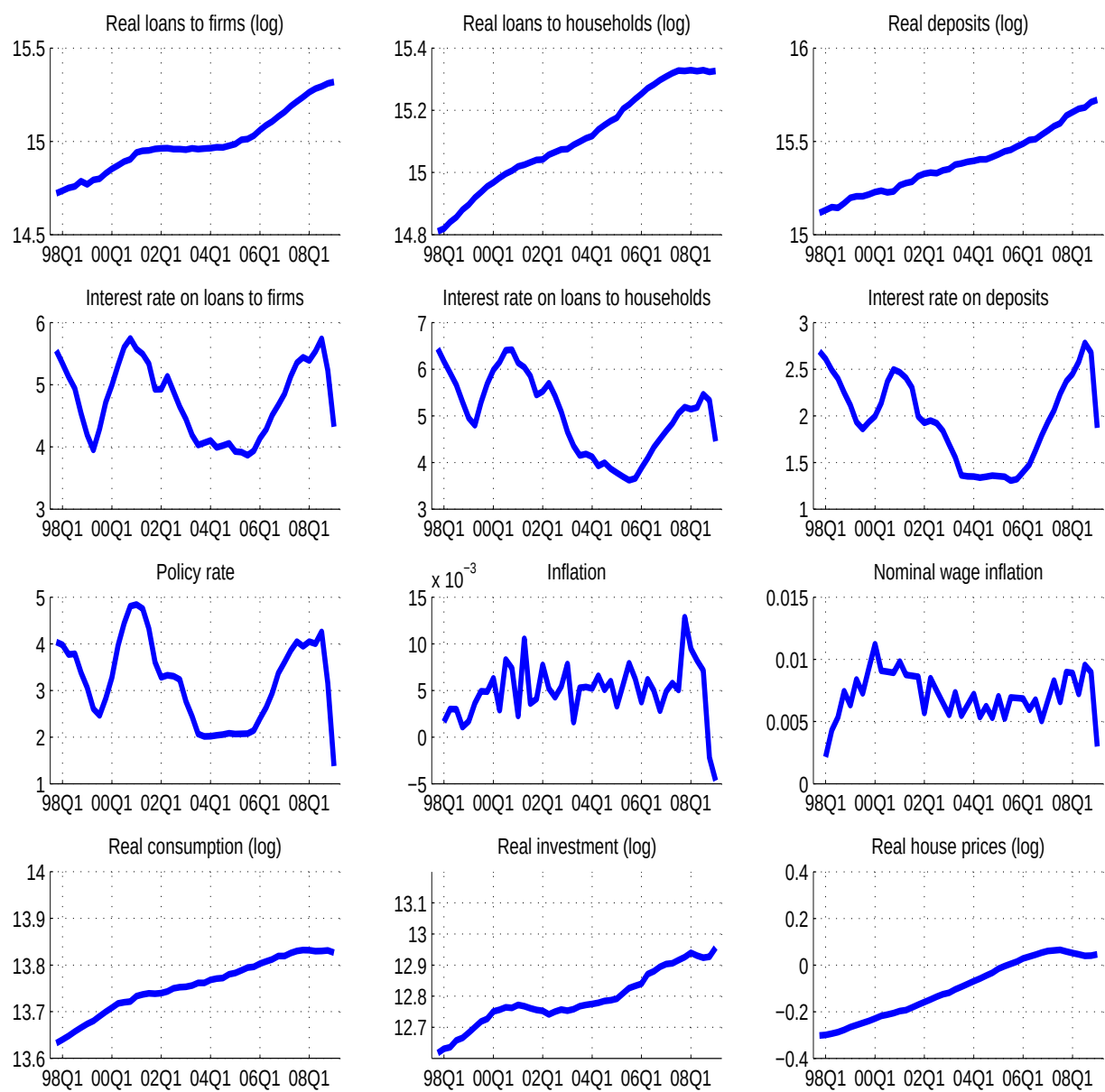


Figure A2. Transformed data

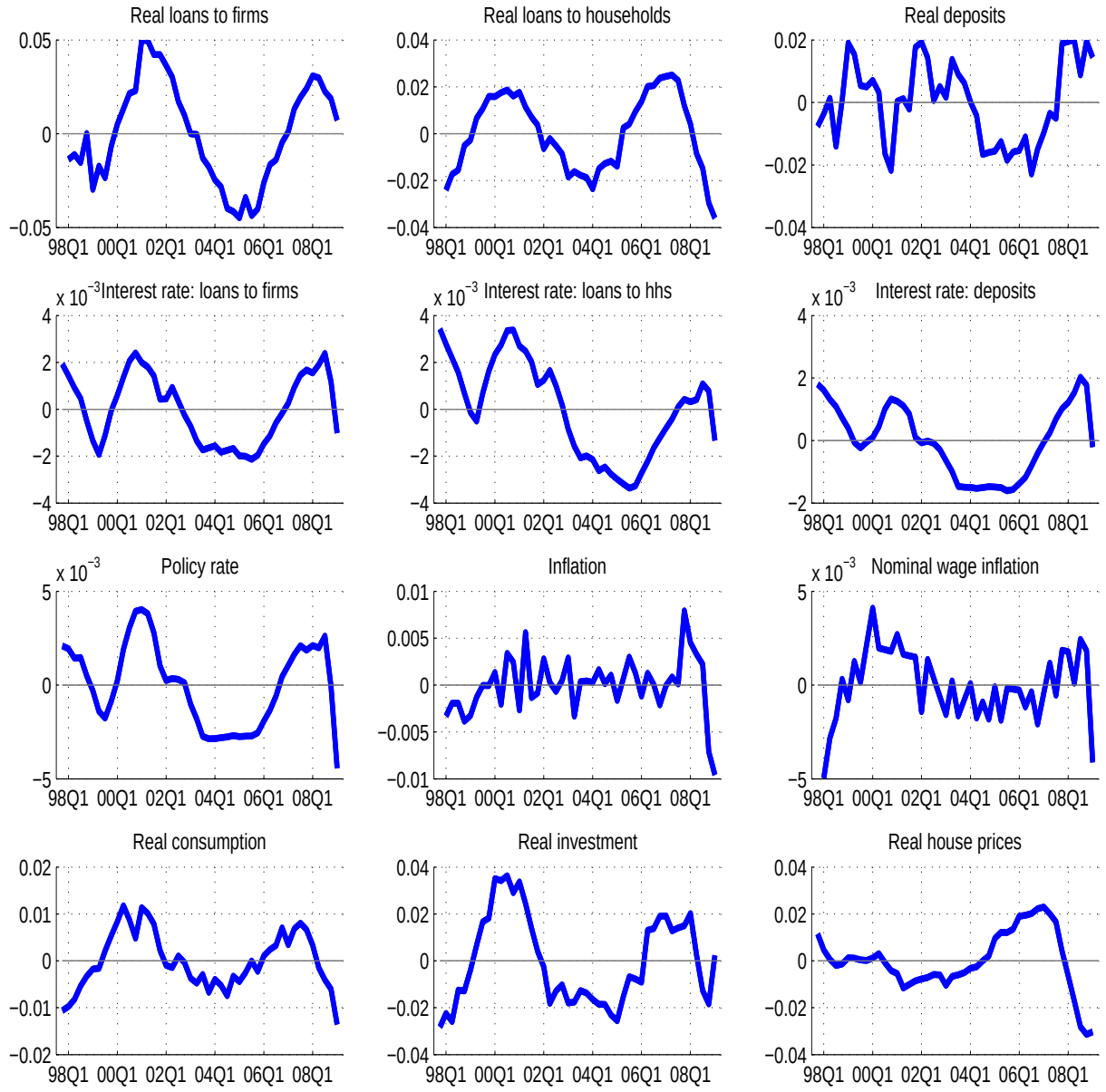
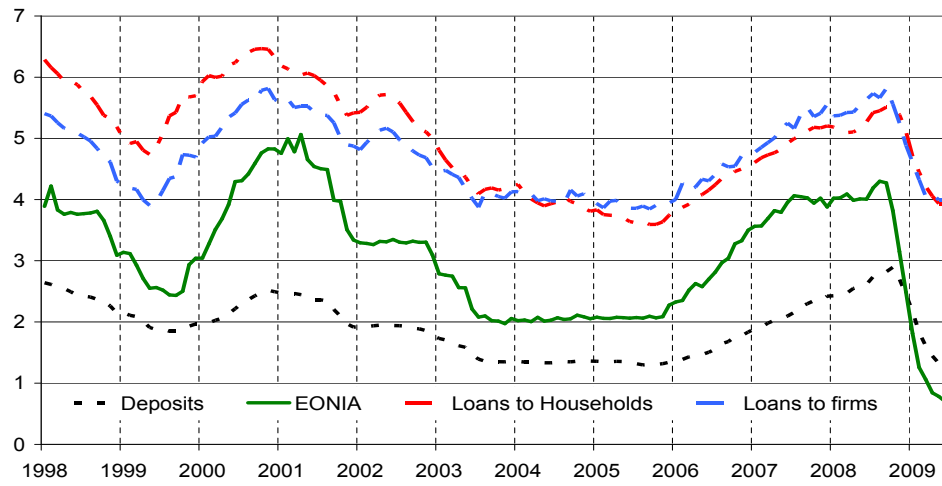


Figure A3. Eonia rate and banks retail rates on loans and deposits (monthly data)



3 Estimation methodology and results

The parameters of the model are estimated using Dynare 4.0. First the mode of the posterior distribution is found using a maximization algorithm, and an approximation of the inverse of the Hessian, at the mode, is computed. Second, we generate 10 chains, each of length 100,000 draws, from the posterior distribution of the parameters using a multivariate normal with covariance matrix proportional to the inverse of the Hessian. Convergence of the algorithm is monitored using the multivariate version of the statistics proposed by Brooks and Gelman (*Journal of Computational and Graphical Statistics*, Vol. 7, No. 4, pp. 434-455, 1998). This statistics is computed automatically in Dynare. We also computed recursive means. Tables A1 and A2 report the summary statistics for the prior and the posterior distributions. Figures A3 to A5 plot the prior and the marginal posterior densities of the parameters of the model.

Table A1. Prior and posterior distribution of the structural parameters

parameter	distr.	prior distribution		posterior distribution			
		mean	std. dev.	mean	2.5 %	median	97.5 %
κ_p	Gamma	50.00	20.00	30.57	10.68	28.65	49.89
κ_w	Gamma	50.00	20.00	102.35	70.29	99.90	133.81
κ_i	Gamma	2.50	1.00	10.26	7.57	10.18	12.81
κ_d	Gamma	10.00	2.50	3.63	2.28	3.50	4.96
κ_{bE}	Gamma	3.00	2.50	9.51	6.60	9.36	12.31
κ_{bH}	Gamma	6.00	2.50	10.22	7.47	10.09	12.88
κ_{Kb}	Gamma	10.00	5.00	11.49	4.03	11.07	18.27
ϕ_π	Gamma	2.00	0.50	2.01	1.72	1.98	2.30
ϕ_R	Beta	0.75	0.10	0.77	0.72	0.77	0.81
ϕ_y	Normal	0.10	0.15	0.35	0.15	0.35	0.55
ι_p	Beta	0.50	0.15	0.17	0.06	0.16	0.28
ι_w	Beta	0.50	0.15	0.28	0.16	0.28	0.39
a^h	Beta	0.50	0.10	0.85	0.81	0.86	0.90

Note: Results based on 10 chains, each with 100,000 draws based on the Metropolis algorithm.

Table A2. Prior and posterior distribution of the structural parameters

parameter	distr.	prior distribution		posterior distribution			
		mean	std.dev	mean	2.5 percent	median	97.5 percent
ρ_z	Beta	0.8	0.10	0.396	0.260	0.394	0.531
ρ_h	Beta	0.8	0.10	0.917	0.858	0.921	0.975
ρ_{mE}	Beta	0.8	0.10	0.892	0.839	0.894	0.945
ρ_{mI}	Beta	0.8	0.10	0.925	0.875	0.929	0.979
ρ_d	Beta	0.8	0.10	0.830	0.739	0.838	0.917
ρ_{bH}	Beta	0.8	0.10	0.808	0.675	0.820	0.949
ρ_{bE}	Beta	0.8	0.10	0.820	0.688	0.834	0.956
ρ_a	Beta	0.8	0.10	0.936	0.899	0.939	0.975
ρ_{qk}	Beta	0.8	0.10	0.543	0.396	0.548	0.694
ρ_y	Beta	0.8	0.10	0.306	0.205	0.305	0.411
ρ_l	Beta	0.8	0.10	0.636	0.511	0.640	0.769
ρ_{Kb}	Beta	0.8	0.10	0.810	0.717	0.813	0.906
σ_z	Inv. gamma	0.01	0.05	0.027	0.019	0.027	0.035
σ_h	Inv. gamma	0.01	0.05	0.076	0.022	0.071	0.129
σ_{mE}	Inv. gamma	0.01	0.05	0.007	0.005	0.007	0.009
σ_{mI}	Inv. gamma	0.01	0.05	0.003	0.003	0.003	0.004
σ_d	Inv. gamma	0.01	0.05	0.033	0.024	0.032	0.043
σ_{bH}	Inv. gamma	0.01	0.05	0.067	0.035	0.066	0.115
σ_{bE}	Inv. gamma	0.01	0.05	0.063	0.034	0.063	0.096
σ_a	Inv. gamma	0.01	0.05	0.006	0.004	0.006	0.007
σ_{qk}	Inv. gamma	0.01	0.05	0.019	0.013	0.019	0.025
σ_R	Inv. gamma	0.01	0.05	0.002	0.001	0.002	0.002
σ_y	Inv. gamma	0.01	0.05	0.634	0.274	0.598	0.985
σ_l	Inv. gamma	0.01	0.05	0.577	0.378	0.561	0.761
σ_{Kb}	Inv. gamma	0.01	0.05	0.031	0.026	0.031	0.037

Note: Results based on 10 chains, each with 100,000 draws based on the Metropolis algorithm.

Figure A4. Prior and posterior marginal distributions

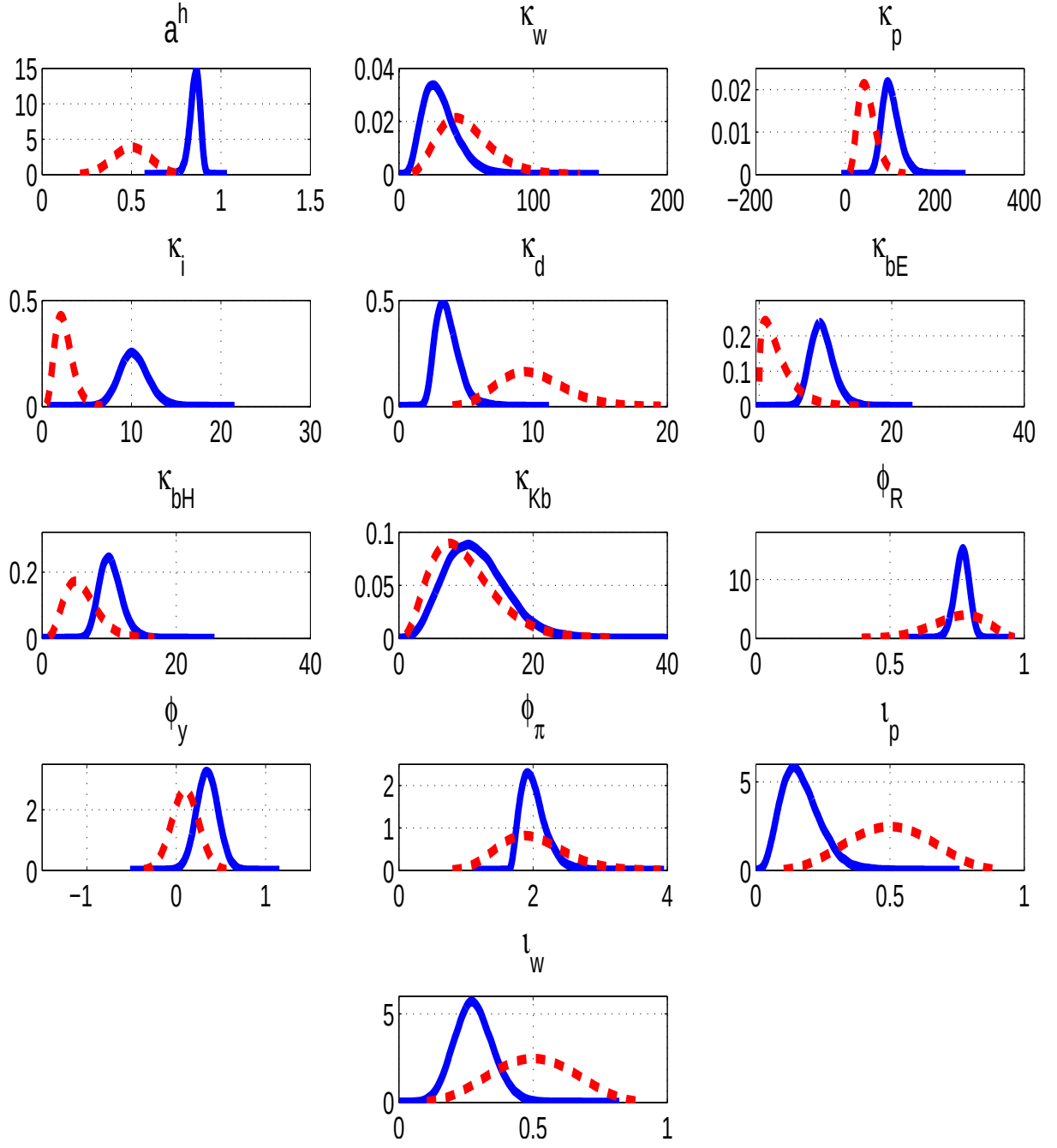


Figure A5. Prior and posterior marginal distributions

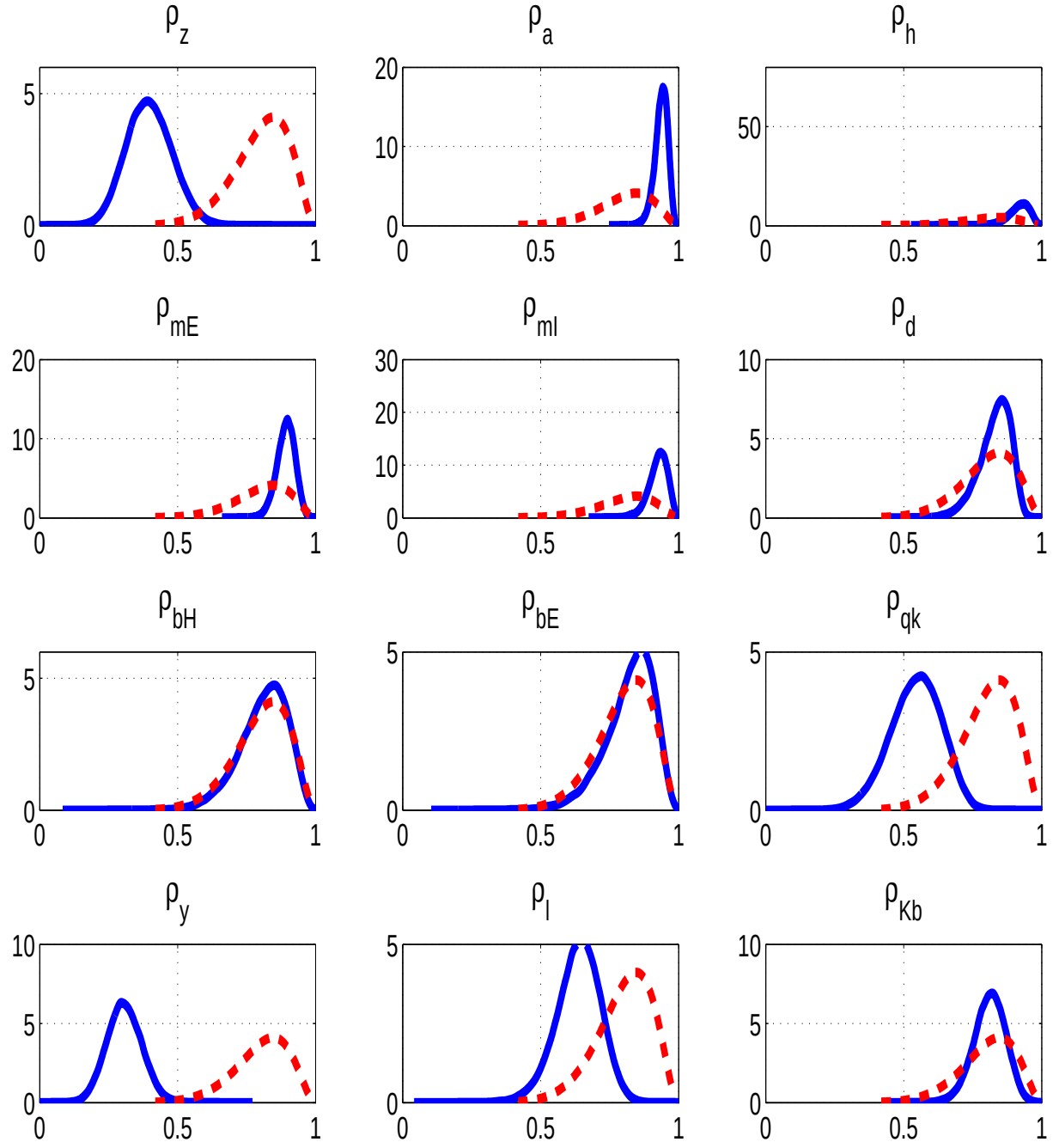
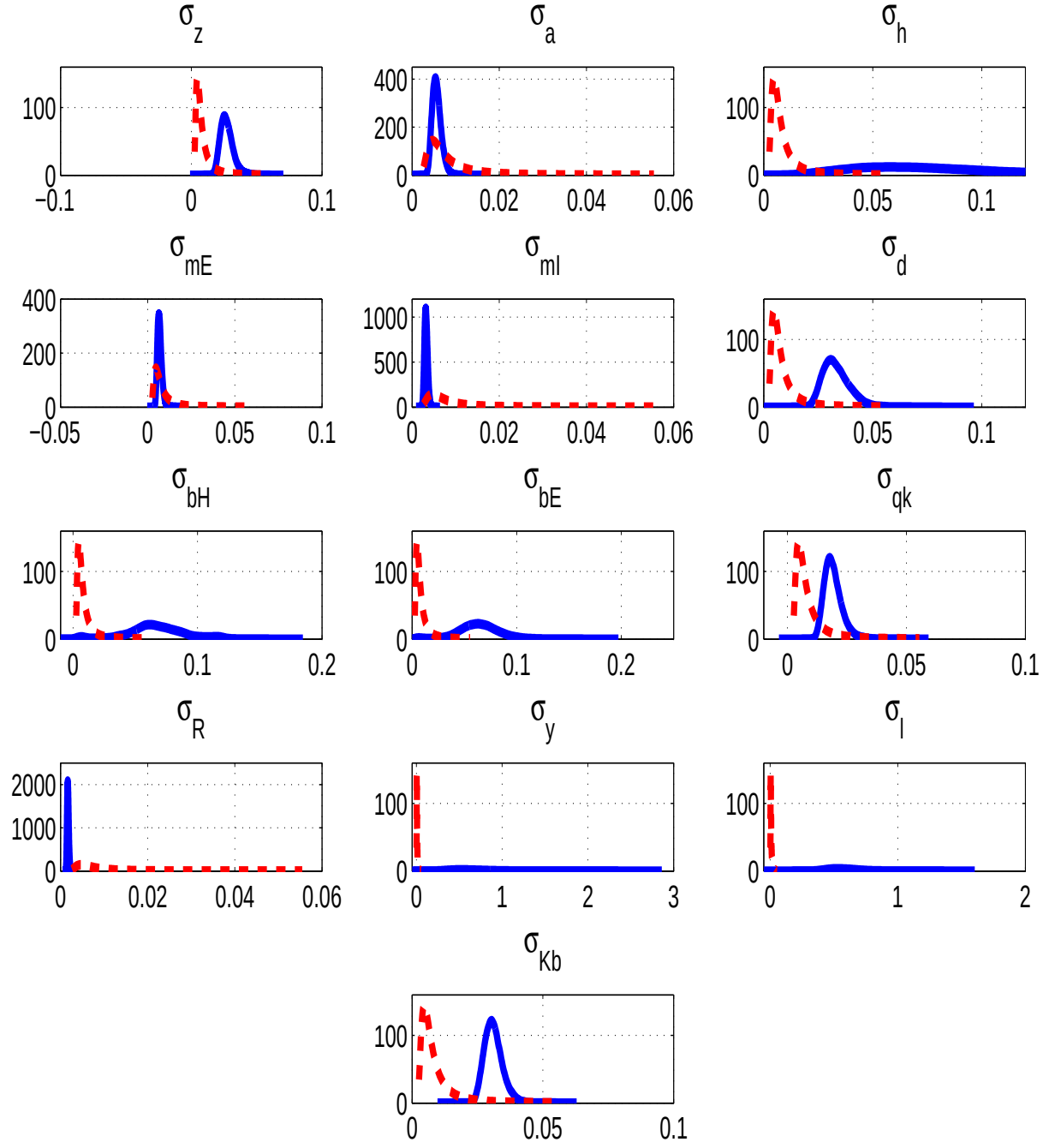


Figure A6. Prior and posterior marginal distributions



4 Convergence diagnostics

In this Section we report some diagnostics to verify that the posterior draws have converged to the true posterior density. Our assessment is based on the multivariate statistics proposed by Brooks and Gelman (*Journal of Computational and Graphical Statistics*, Vol. 7, No. 4, pp. 434-455, 1998), on recursive means of the parameters and on potential reduction scale factors. Figure A7 reports the multivariate convergence diagnostics based on the the eighty percent interval and on the second and third moments; the solid lines display the statistics based on the total number of draws (1,000,000) while the dashed lines the same statistics based on within-means. Figures A8 to A11 report the recursive means computed with all the draws from the ten Markov chains generated by the Metropolis algorithm. Tables A3 and A4 report the potential reduction scale factors, a measure of how much the precision of the posterior mean can be expected to improve if we were to continue the MCMC runs indefinitely: our resulting statistics are all below 1.1 and are therefore acceptable, according to Gelman, Carlin, Stern and Rubin (*Bayesian Data Analysis*, 2nd edition, Chapman & Hall, New York, 2004).

Figure A7. Multivariate convergence diagnostics

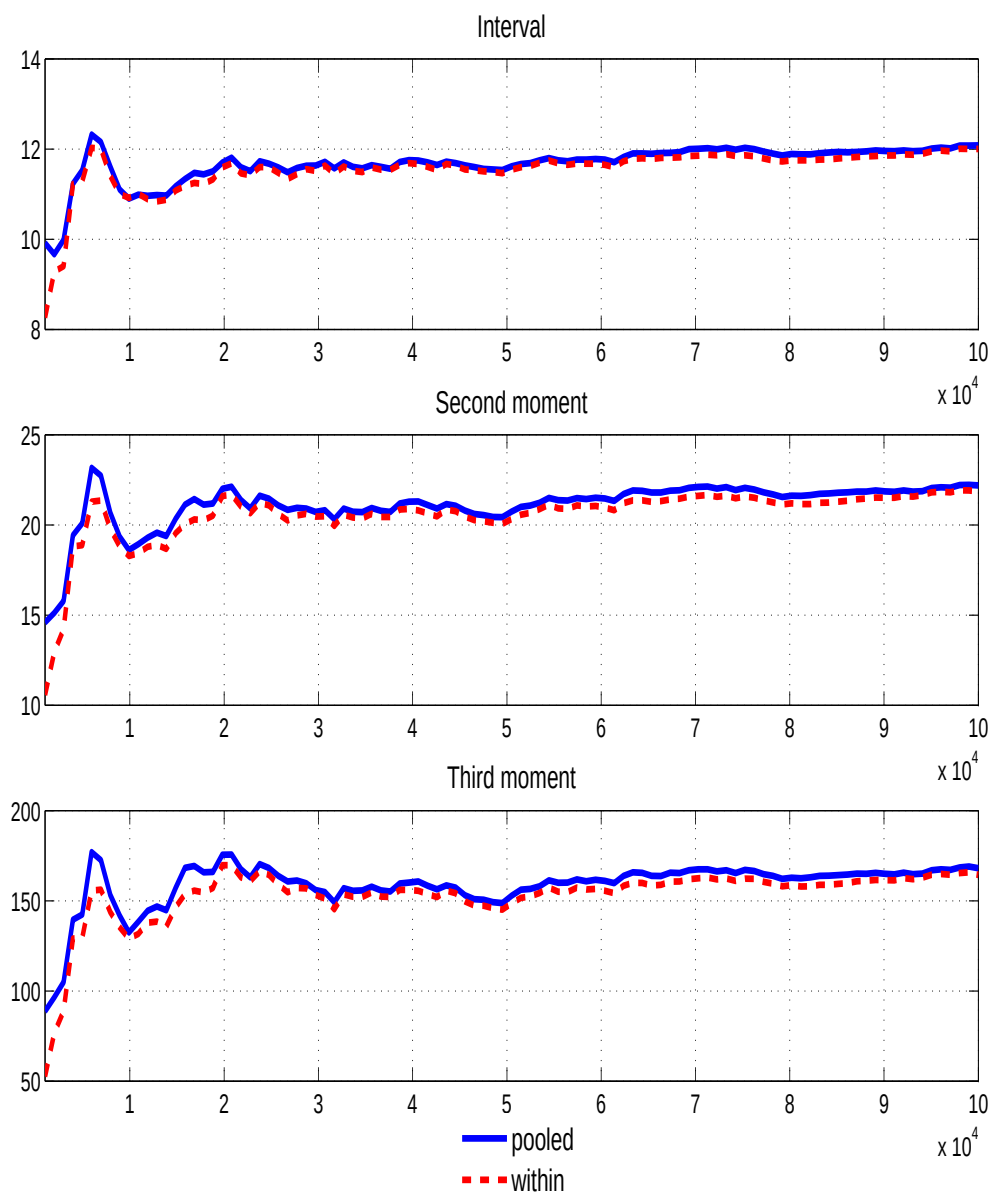
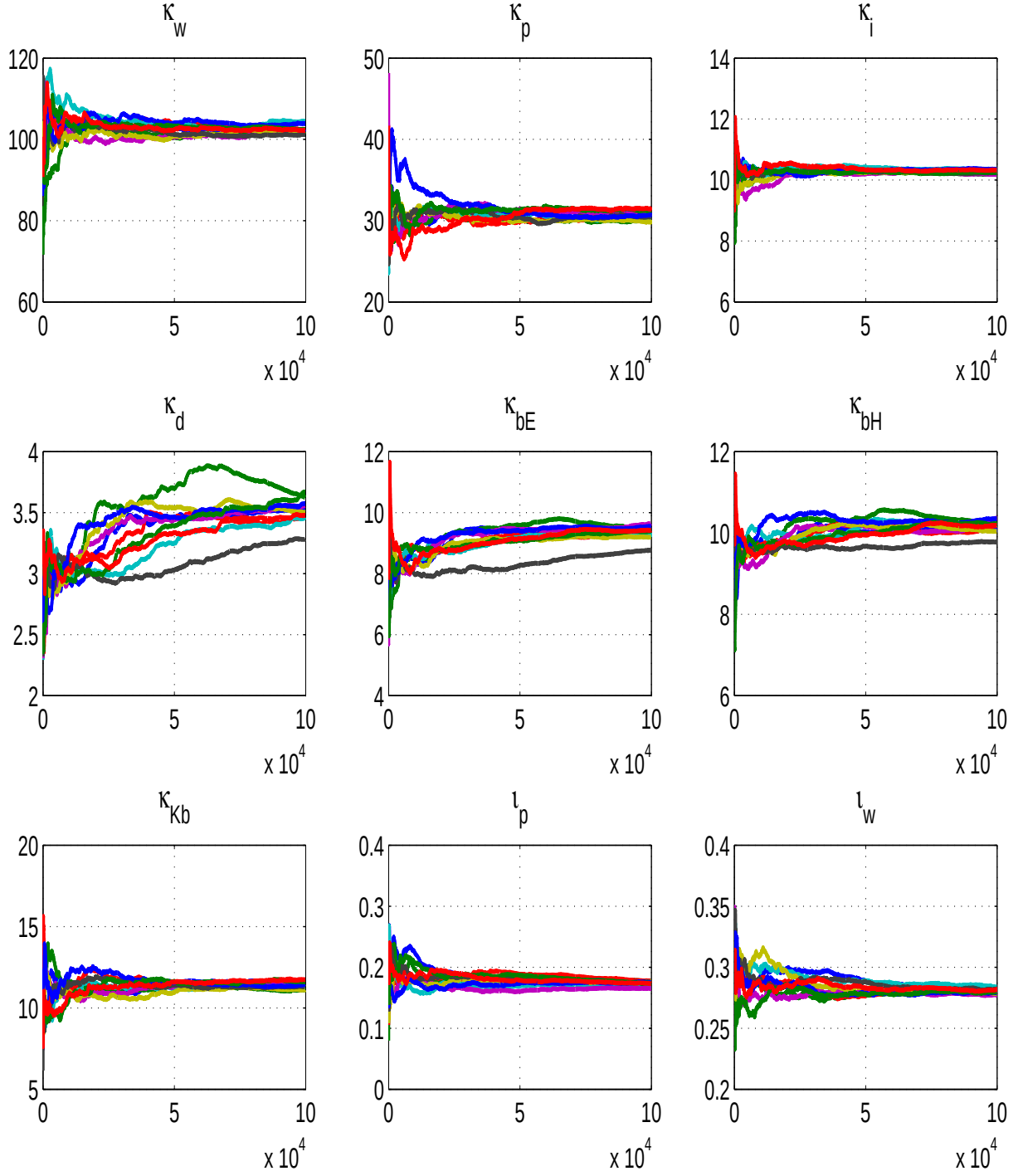
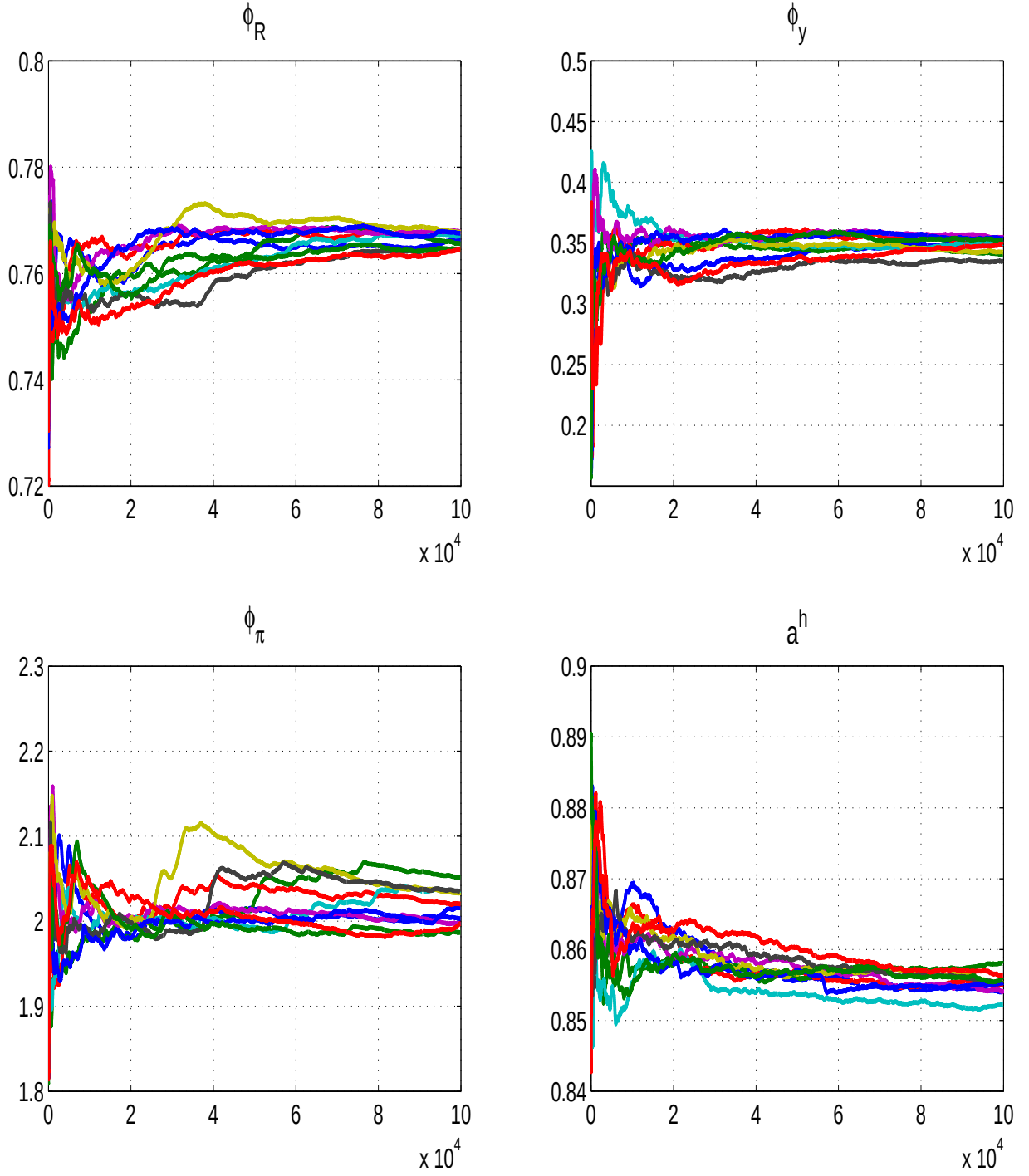


Figure A8. Recursive means



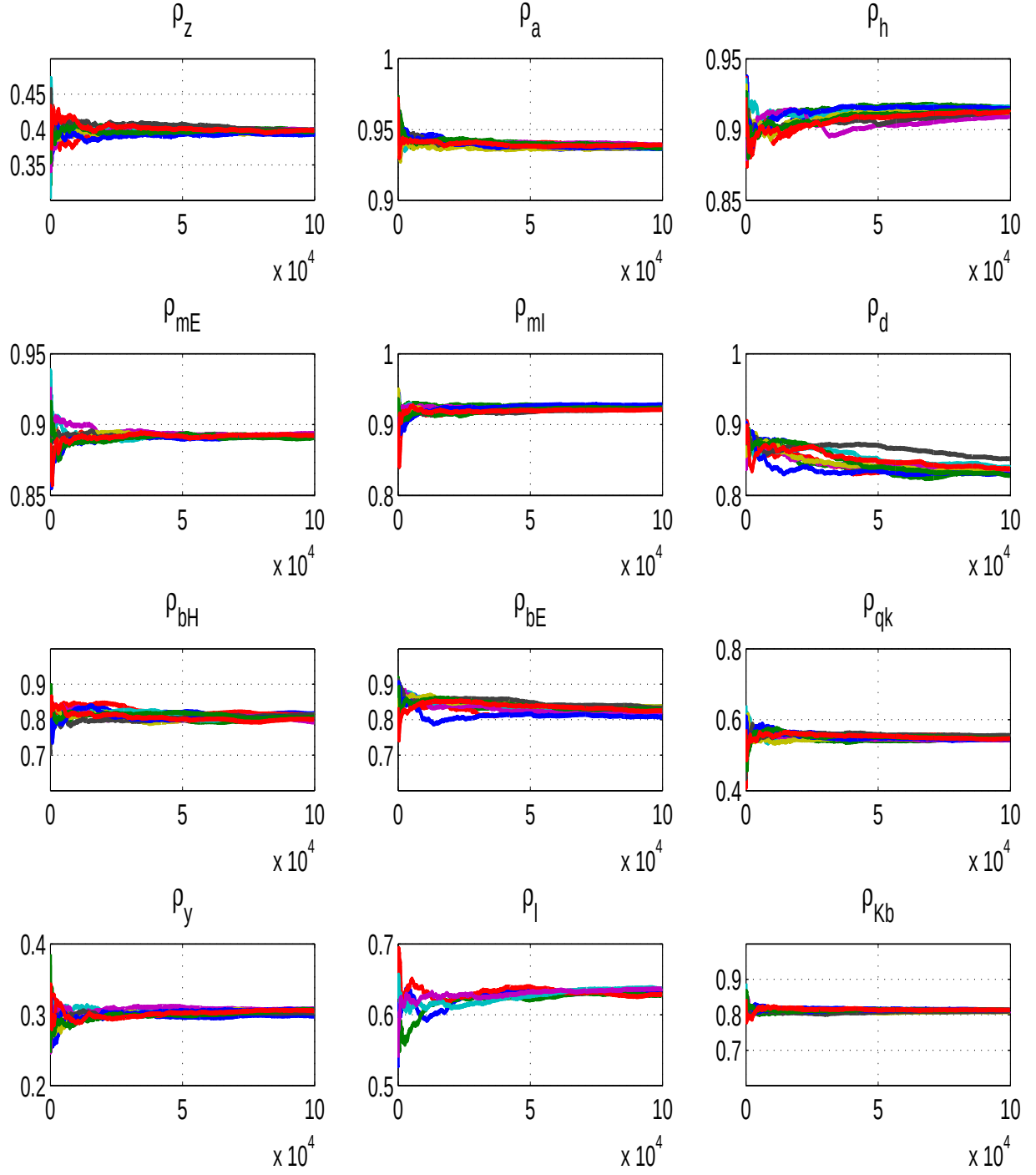
The graphs report the recursive means based on ten chains each of length 100,000 draws.

Figure A9. Recursive means



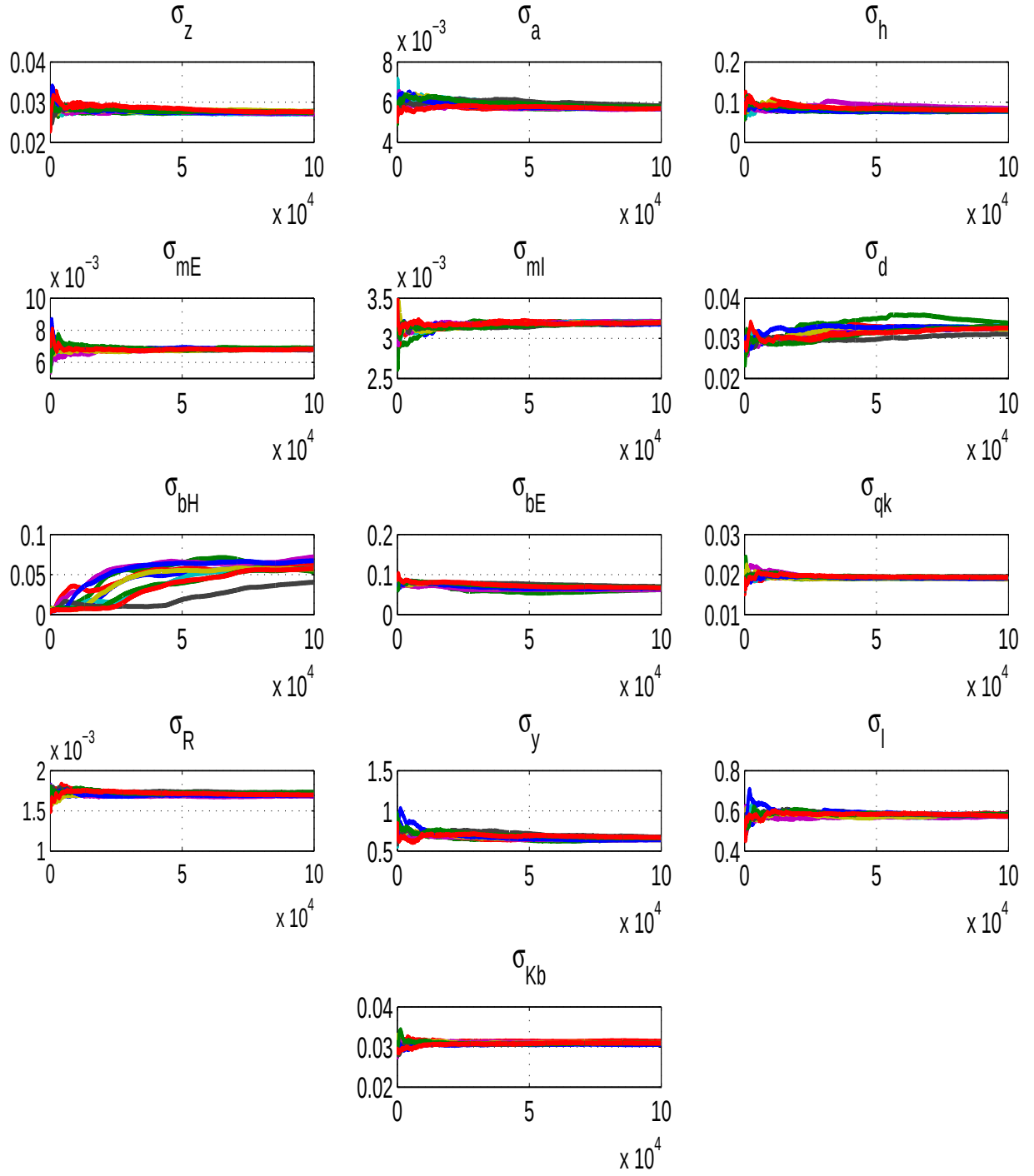
The graphs report the recursive means based on ten chains each of length 100,000 draws.

Figure A10. Recursive means



The graphs report the recursive means based on ten chains each of length 100,000 draws.

Figure A11. Recursive means



The graphs report the recursive means based on ten chains each of length 100,000 draws.

Table A3. Potential reduction scale (PRS) factors
of the structural parameters

parameter	PRS factor
κ_p	1.0098
κ_w	1.0136
κ_i	1.0077
κ_d	1.0716
κ_{bE}	1.0848
κ_{bH}	1.0525
κ_{Kb}	1.0148
ϕ_π	1.0565
ϕ_R	1.0150
ϕ_y	1.0137
ι_p	1.0138
ι_w	1.0035
a^h	1.0147

Note: Results based on 10 chains, each with 100,000 draws.

Table A4. Potential reduction scale (PRS) factors
of the structural parameters

parameter	PRS factor
ρ_z	1.0031
ρ_h	1.0140
ρ_{mE}	1.0035
ρ_{mI}	1.0151
ρ_d	1.0605
ρ_{bH}	1.0326
ρ_{bE}	1.0539
ρ_a	1.0112
ρ_{qk}	1.0089
ρ_y	1.0119
ρ_l	1.0276
ρ_{Kb}	1.0051
σ_z	1.0087
σ_h	1.0209
σ_{mE}	1.0051
σ_{mI}	1.0053
σ_d	1.0686
σ_{bH}	1.3898
σ_{bE}	1.0825
σ_a	1.0143
σ_{qk}	1.0100
σ_R	1.0296
σ_y	1.0139
σ_l	1.0065
σ_{Kb}	1.0179

Note: Results based on 10 chains, each with 100,000 draws.

5 Robustness analysis

5.1 Data transformation and features of the model

In this Section we report the results of some robustness exercises. In particular, we estimate the benchmark model with data that are linearly-detrended in order to extract their business cycle component (LT-data). We use this filter for the variables with a trend while those that are stationary are simply de-meanned as in the benchmark case. We then focus on a version of the model without financial shocks (No financial shocks), a version with flexible rates but still with monopolistically competitive banks (Flexible rates) and finally a version with sticky bank rates but no bank capital (No bank capital). We estimate these three models with HP-detrended data as done for the benchmark model. Tables A5 and A6 compare the median values of the parameters across specifications of the model and data while Table A7 reports the marginal densities of the benchmark model, the version without financial shocks, the one with flexible bank rates and finally a version without bank capital.

Table A5. Posterior median of alternative models

parameter	Benchmark	LT data	No financial shocks	Flexible rates	No bank capital
κ_p	28.65	28.68	31.55	66.18	27.70
κ_w	99.90	104.70	64.95	78.17	106.02
κ_i	10.18	10.26	4.73	10.01	10.25
κ_d	3.50	3.81	10.63	-	3.90
κ_{bE}	9.36	10.03	1.15	-	9.96
κ_{bH}	10.09	10.46	9.55	-	10.65
κ_{Kb}	11.07	10.11	38.19	8.99	-
ϕ_π	1.98	2.05	2.17	2.09	2.02
ϕ_R	0.77	0.78	0.76	0.87	0.78
ϕ_Y	0.35	0.31	0.40	0.28	0.33
ι_p	0.16	0.17	0.30	0.22	0.16
ι_w	0.28	0.27	0.29	0.38	0.27
a^h	0.86	0.82	0.69	0.84	0.85

Note: LT stands for the benchmark model with linearly-detrended data. Results based on 10 chains, each with 100,000 draws based on the Metropolis algorithm.

Table A6. Posterior median of alternative models

parameter	Benchmark	LT data	No financial shocks	Flexible rates	No bank capital
ρ_z	0.394	0.453	0.597	0.347	0.388
ρ_h	0.921	0.979	0.805	0.902	0.925
ρ_{mE}	0.894	0.934	-	0.893	0.900
ρ_{mI}	0.929	0.946	-	0.843	0.932
ρ_d	0.838	0.846	-	0.944	0.801
ρ_{bH}	0.820	0.823	-	0.909	0.794
ρ_{bE}	0.834	0.851	-	0.890	0.844
ρ_a	0.939	0.939	0.984	0.668	0.945
ρ_{qk}	0.548	0.542	0.318	0.528	0.552
ρ_y	0.305	0.306	0.773	0.909	0.311
ρ_l	0.640	0.630	0.787	0.814	0.639
ρ_{Kb}	0.813	0.853	-	0.877	0.799
σ_z	0.027	0.025	0.014	0.024	0.026
σ_h	0.071	0.029	0.129	0.066	0.068
σ_{mE}	0.007	0.007	-	0.007	0.007
σ_{mI}	0.003	0.004	-	0.003	0.003
σ_d	0.032	0.033	-	0.032	0.035
σ_{bH}	0.066	0.064	-	0.120	0.075
σ_{bE}	0.063	0.069	-	0.111	0.062
σ_a	0.006	0.006	0.005	0.014	0.005
σ_{qk}	0.019	0.019	0.015	0.019	0.018
σ_R	0.002	0.002	0.002	0.002	0.002
σ_y	0.598	0.609	0.379	0.404	0.549
σ_l	0.561	0.579	0.297	0.007	0.606
σ_{Kb}	0.031	0.033	-	0.030	0.032

Note: LT stands for the benchmark model with linearly-detrended data. Results based on 10 chains, each with 100,000 draws based on the Metropolis algorithm.

Table A7. Comparison of models: marginal density

benchmark	No financial shocks	Flexible rates	No bank capital
2311.47	2017.97	2261.86	2307.62

Note: All models are estimated with HP-detrended data. Results based on 10 chains, each with 100,000 draws based on the Metropolis algorithm.

5.2 Sensitivity with respect to the prior distribution

In this Section we report the results of the Bayesian estimation (posterior median values) when the standard deviations of the prior distributions are increased by 50 percent (Tables A8 and A9). Figures A12 and A13 plot the marginal prior and posterior distributions in the case of the benchmark specification of the prior and in the “sensitivity” case.

Table A8. Posterior median of benchmark model: prior sensitivity

benchmark				sensitivity w.r.t prior std. dev.		
parameter	mean	std. dev.	median	mean	std. dev.	median
κ_p	50.0	20.0	28.65	50.0	30.0	27.83
κ_w	50.0	20.0	99.90	50.0	30.0	129.59
κ_i	2.5	1.0	10.18	2.5	1.5	14.60
κ_d	10.0	2.5	3.50	10.0	3.75	2.53
κ_{bE}	3.0	2.5	9.36	3.0	3.75	9.69
κ_{bH}	6.0	2.5	10.09	6.0	3.75	9.94
κ_{Kb}	10.0	5.0	11.07	10.0	7.5	13.68
ϕ_π	2.0	0.5	1.98	2.0	0.75	2.00
ϕ_R	0.75	0.10	0.77	0.75	0.225	0.76
ϕ_Y	0.10	0.15	0.35	0.10	0.225	0.51
ι_p	0.50	0.15	0.16	0.50	0.225	0.09
ι_w	0.50	0.15	0.28	0.50	0.225	0.24
a^h	0.50	0.10	0.86	0.50	0.15	0.88

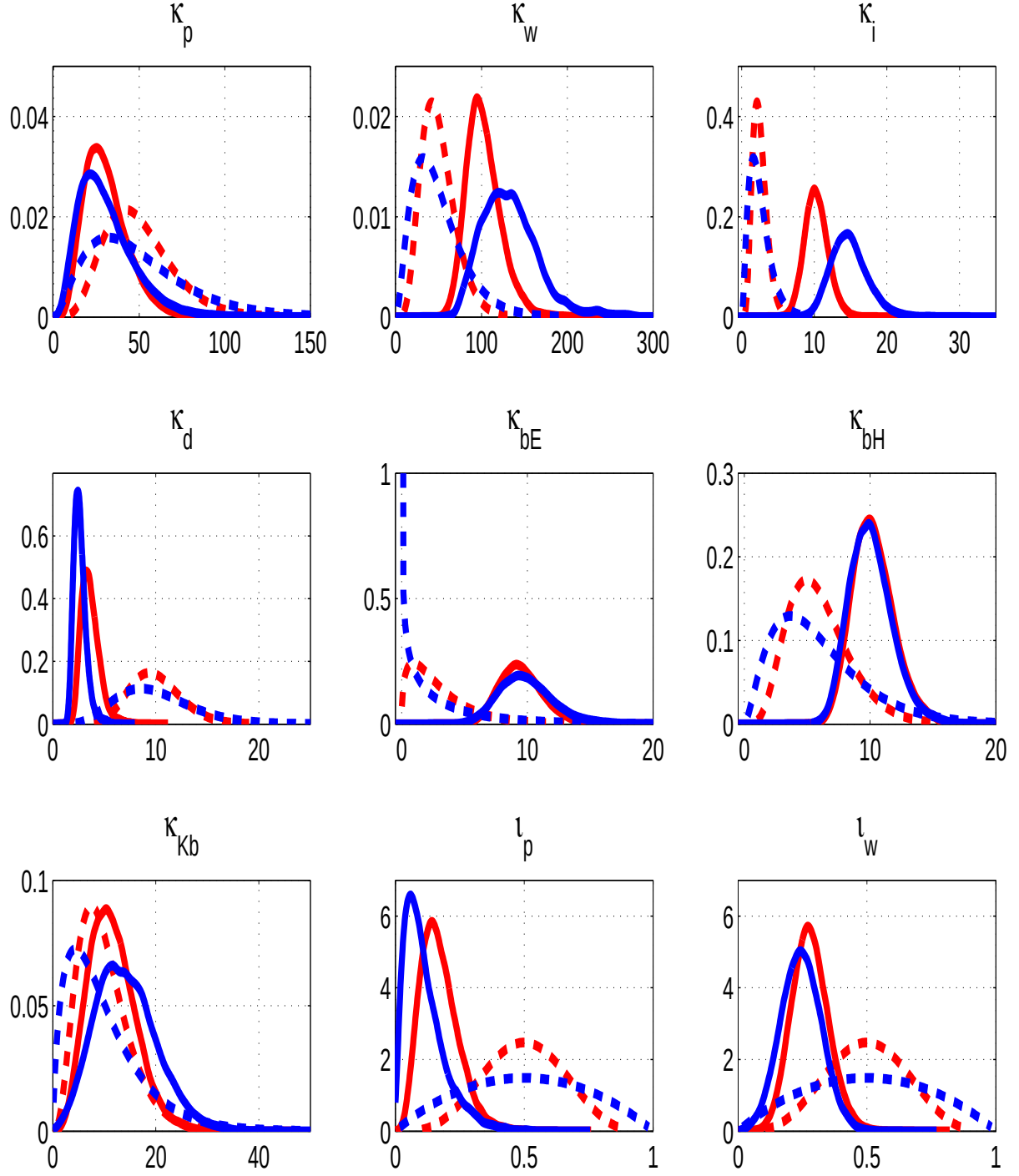
Note: All models are estimated with HP-detrended data. Results based on 10 chains, each with 100,000 draws based on the Metropolis algorithm.

Table A9. Posterior median of benchmark model: prior sensitivity

parameter	benchmark			sensitivity w.r.t prior std. dev.		
	mean	std. dev.	median	mean	std. dev.	median
ρ_z	0.8	0.100	0.394	0.8	0.150	0.227
ρ_h	0.8	0.100	0.939	0.8	0.150	0.938
ρ_{mE}	0.8	0.100	0.897	0.8	0.150	0.900
ρ_{mI}	0.8	0.100	0.953	0.8	0.150	0.956
ρ_d	0.8	0.100	0.908	0.8	0.150	0.889
ρ_{bH}	0.8	0.100	0.893	0.8	0.150	0.870
ρ_{bE}	0.8	0.100	0.849	0.8	0.150	0.838
ρ_a	0.8	0.100	0.939	0.8	0.150	0.939
ρ_{qk}	0.8	0.100	0.372	0.8	0.150	0.355
ρ_y	0.8	0.100	0.229	0.8	0.150	0.220
ρ_l	0.8	0.100	0.522	0.8	0.150	0.556
ρ_{Kb}	0.8	0.100	0.859	0.8	0.150	0.849
σ_z	0.01	0.050	0.027	0.01	0.075	0.027
σ_h	0.01	0.050	0.071	0.01	0.075	0.052
σ_{mE}	0.01	0.050	0.007	0.01	0.075	0.007
σ_{mI}	0.01	0.050	0.003	0.01	0.075	0.003
σ_d	0.01	0.050	0.032	0.01	0.075	0.027
σ_{bH}	0.01	0.050	0.066	0.01	0.075	0.055
σ_{bE}	0.01	0.050	0.063	0.01	0.075	0.070
σ_a	0.01	0.050	0.006	0.01	0.075	0.005
σ_{qk}	0.01	0.050	0.019	0.01	0.075	0.013
σ_R	0.01	0.050	0.002	0.01	0.075	0.002
σ_y	0.01	0.050	0.598	0.01	0.075	0.600
σ_l	0.01	0.050	0.561	0.01	0.075	0.733
σ_{Kb}	0.01	0.050	0.031	0.01	0.075	0.031

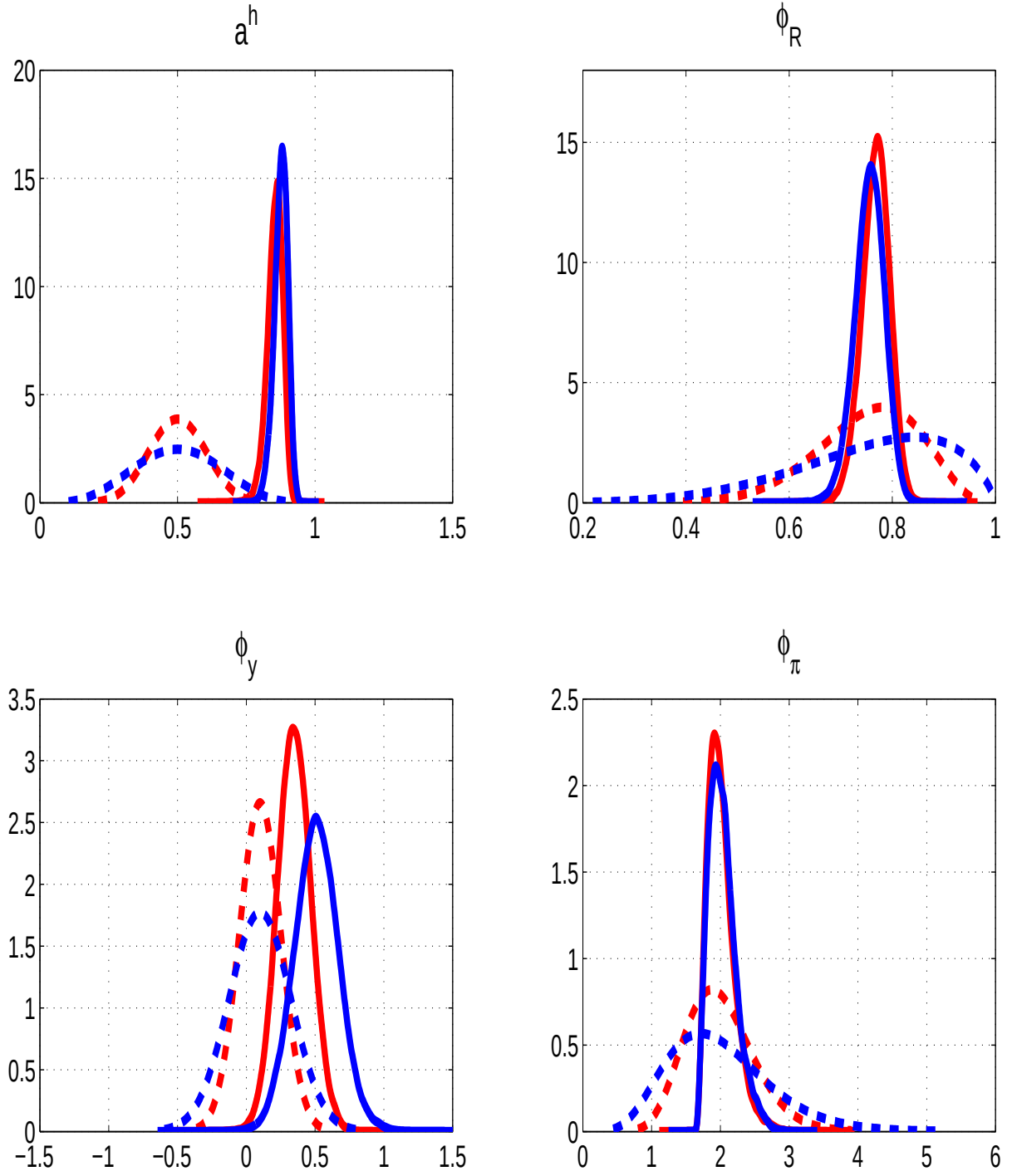
Note: All models are estimated with HP-detrended data. Results based on 10 chains, each with 100,000 draws based on the Metropolis algorithm.

Figure A12. Prior and posterior marginal distributions: the role of the prior



Note: dashed red: benchmark prior; solid red: benchmark posterior; dashed blue: "sensitivity" prior; solid blue: "sensitivity" posterior.

Figure A13. Prior and posterior marginal distributions: the role of the prior



Note: dashed red: benchmark prior; solid red: benchmark posterior; dashed blue: “sensitivity” prior; solid blue: “sensitivity” posterior.